

I. DETAIL OF HYPERPARAMETERS

In our study, hyperparameter tuning was performed manually through empirical testing. For each hyperparameter, we conducted trial-and-error searches within commonly used ranges. The parameters, candidate range, and the final selected values are summarized in Table S1.

TABLE S1
HYPERPARAMETER CANDIDATES AND SELECTED VALUES

Parameter	Candidate Range	Selected Value
Learning rate	0.0001 - 0.001	0.001
Batch size	16 - 64	32
Hidden dimension	64 - 512	128
Number of Transformer layers	4 - 10	8
Number of GAT layers	2 - 6	4
Number of training epochs	100 - 300	200
Early stopping patience	10 - 30	20
Dropout rate	0.1 - 0.5	0.2

II. DETAIL OF EVALUATION METRICS

Our study evaluates model performance using both example-based and label-based metrics to ensure a comprehensive and fair comparison across different methods. Six example-based metrics including $ACC_{example}$, Average Precision, Hamming Loss, Ranking Loss, One-error, and Coverage, which are defined as:

$$ACC_{example} = \frac{1}{N} \sum_{i=1}^N \frac{|Y_i \cap P_i|}{|Y_i \cup P_i|} \quad (1)$$

$$Average\ Precision = \frac{1}{L} \sum_{j=1}^L \sum_{k=1}^N P_{j,k} (R_{j,k} - R_{j,k-1}) \quad (2)$$

$$Hamming\ Loss = \frac{1}{N \cdot L} \sum_{i=1}^N \sum_{j=1}^L 1(y_i[j] \neq \hat{y}_i[j]) \quad (3)$$

$$Ranking\ Loss = \frac{1}{N} \sum_{i=1}^N \frac{1}{|P_i| \cdot |\bar{P}_i|} \sum_{j \in P_i} \sum_{k \in \bar{P}_i} 1(f_i(j) \leq f_i(k)) \quad (4)$$

$$One - error = \frac{1}{N} \sum_{i=1}^N 1(\argmax_{j \in \{1,2,\dots,L\}} f_i(j)) \notin P_i \quad (5)$$

$$Coverage = \frac{1}{N} \sum_{i=1}^N (\max_{j \in Y_i} (rank_i(j)) - 1) \quad (6)$$

where N is the total number of samples, L is the total number of labels, P_i is the true label set for the i -th sample, Y_i is the predicted label set for the i -th sample, $y_i[j]$ is the true binary value of the i -th sample on the j -th label, $\hat{y}_i[j]$ is the predicted binary value of the i -th sample on the j -th label, $f_i(j)$ denotes the predicted score for the i -th sample on label j , $rank_i(j)$ denotes the rank of label j in the order of predicted scores of the i -th sample, and $1(\cdot)$ denotes the indicator function. Four label-based metrics, including average F1-score (AvgF1), micro precision (MiP), micro recall (MiR), macro precision

(MaP), macro recall (MaR), and average area under the curve (AvgAUC), were utilized and are defined as follows:

$$AvgF1 = \frac{1}{L} \sum_{l=1}^L \frac{2 \cdot TP_l}{2 \cdot TP_l + FP_l + FN_l} \quad (7)$$

$$MiP = \frac{\sum_{l=1}^L TP_l}{\sum_{l=1}^L (TP_l + FP_l)} \quad (8)$$

$$MiR = \frac{\sum_{l=1}^L TP_l}{\sum_{l=1}^L (TP_l + FN_l)} \quad (9)$$

$$MaP = \frac{1}{L} \sum_{l=1}^L \frac{TP_l}{TP_l + FP_l} \quad (10)$$

$$MaR = \frac{1}{L} \sum_{l=1}^L \frac{TP_l}{TP_l + FN_l} \quad (11)$$

$$AvgAUC = \frac{1}{L} \sum_{l=1}^L \frac{\sum_{i \in P_l} \sum_{j \in N_l} 1(f_i(l) > f_j(l)) + 0.5 \cdot 1(f_i(l) = f_j(l))}{|P_l| \cdot |N_l|} \quad (12)$$

$$AvgMCC = \frac{1}{L} \sum_{l=1}^L \frac{TP_l \cdot TN_l - FP_l \cdot FN_l}{\sqrt{(TP_l + FP_l)(TP_l + FN_l)(TN_l + FP_l)(TN_l + FN_l)} + \varepsilon} \quad (13)$$

where L represents the total number of labels, TP_l represents the number of true positive samples for the l -th label, FP_l represents the number of false positive samples for the l -th label, TN_l represents the number of true negative samples for the l -th label, FN_l represents the number of false negative samples for the l -th label, P_l , N_l represent the sets of positive and negative samples for the l -th label, respectively, and ε is a bias term introduced to prevent division by zero.

III. CROSS DATASET EVALUATION FOR GENERALIZABILITY ASSESSMENT

To further validate the robustness of LncTracker, we conducted cross-dataset evaluations using benchmark datasets from prior tools with publicly available data, these are LncLocator 2.0 [1], DeepLncLoc [2], GraphLncLoc [3], and LncLocFormer [4]. Each dataset's task formulation was preserved—multi-label for LncLocFormer and single-label for DeepLncLoc and GraphLncLoc. Accordingly, we replaced the Sigmoid activation function in LncTracker's output layer with Softmax to normalize the output and facilitate clearer evaluation of the model's predicted probabilities for each class. Specifically, we excluded multi-label-specific metrics such as Average Precision, Hamming Loss, Ranking Loss, and Coverage [5], and adopted Macro Precision and Macro Recall in place of Micro-based metrics, as the latter yield identical values in single-label settings.

TABLE S2
EXAMPLE-BASED METRICS COMPARISON ON EXTERNAL PUBLIC DATASETS

Predictor	Data Source	Dataset size(train/test)	$ACC_{example}$	Average precision	Hamming loss	Ranking loss	One-error	Coverage
LncLocator 2.0	lncATLAS	490/122	0.513	-	-	-	0.587	-
LncTracker	lncATLAS	490/122	0.526	-	-	-	0.474	-
DeepLncLoc	RNAlocate 1.0	686/171	0.514	-	-	-	0.485	-
LncTracker	RNAlocate 1.0	686/171	0.559	-	-	-	0.441	-
GraphLncLoc	RNAlocate 1.0	616/153	0.519	-	-	-	0.481	-
LncTracker	RNAlocate 1.0	616/153	0.628	-	-	-	0.372	-
LncLocFormer	RNAlocate 2.0	649/162	0.580	0.563	0.232	0.392	0.634	2.659
LncTracker	RNAlocate 3.0	649/162	0.615	0.596	0.214	0.338	0.623	2.480

TABLE S3
LABEL-BASED METRICS COMPARISON ON EXTERNAL PUBLIC DATASETS

Predictor	Data Source	Dataset size(train/test)	AvgF1	MiP	MiR	MaP	MaR	AvgAUC	AvgMCC
LncLocator 2.0	lncATLAS	490/122	0.522	-	-	0.609	0.413	0.612	0.128
LncTracker	lncATLAS	490/122	0.526	-	-	0.550	0.467	0.720	0.144
DeepLncLoc	RNAlocate 1.0	686/171	0.514	-	-	0.461	0.351	0.700	0.237
LncTracker	RNAlocate 1.0	686/171	0.558	-	-	0.518	0.358	0.762	0.233
GraphLncLoc	RNAlocate 1.0	616/153	0.519	-	-	0.508	0.425	0.655	0.268
LncTracker	RNAlocate 1.0	616/153	0.628	-	-	0.416	0.471	0.769	0.201
LncLocFormer	RNAlocate 2.0	649/162	0.689	0.701	0.591	-	-	0.549	0.033
LncTracker	RNAlocate 3.0	649/162	0.731	0.690	0.738	-	-	0.638	0.040

TABLE S4
EXAMPLE-BASED METRICS COMPARISON ON EXTREME LENGTH DATASET (SEQUENCE LENGTH LESS THAN 500 OR MORE THAN 5000)

Predictor	Dataset size (train/test)	$ACC_{example}$	Average precision	Hamming loss	Ranking loss	One-error	Coverage
LncLocator	3560 / 510	0.394	0.569	0.408	0.592	0.951	5.907
DeepLncLoc	3560 / 510	0.530	0.556	0.298	0.340	0.827	5.364
LncLocator 2.0	3560 / 510	0.531	0.564	0.249	0.366	0.860	5.266
GraphLncLoc	3560 / 510	0.464	0.553	0.280	0.432	0.871	5.430
LncLocFormer	3560 / 510	0.542	0.577	0.250	0.390	0.834	5.159
LncTracker	3560 / 510	0.545	0.584	0.239	0.336	0.875	5.257

TABLE S5
LABEL-BASED METRICS COMPARISON ON EXTREME LENGTH DATASET (SEQUENCE LENGTH LESS THAN 500 OR MORE THAN 5000)

Predictor	Dataset size (train/test)	AvgF1	MiP	MiR	AvgAUC	AvgMCC
LncLocator	3560 / 510	0.595	0.500	0.591	0.519	0.022
DeepLncLoc	3560 / 510	0.704	0.622	0.488	0.516	0
LncLocator 2.0	3560 / 510	0.707	0.751	0.566	0.604	0.047
GraphLncLoc	3560 / 510	0.657	0.750	0.471	0.570	0
LncLocFormer	3560 / 510	0.716	0.720	0.636	0.637	0.072
LncTracker	3560 / 510	0.721	0.761	0.609	0.684	0.131

Results in Table S2 and Table S3 show that LncTracker consistently outperforms prior methods across example-based and label-based metrics. Notably, it achieves the highest scores in average F1, Micro/Macro Recall, and average AUC. Interestingly, average MCC was higher in single-label tasks, likely due to reduced label complexity compared to multi-label scenarios.

IV. ROBUSTNESS EVALUATION ON ATYPICALLY SHORT AND LONG SEQUENCES

To further assess the robustness and generalizability of LncTracker under extreme input conditions, we conducted an additional evaluation focused on RNA sequences of atypical lengths. Specifically, the original benchmark dataset was partitioned based on sequence length: all sequences shorter than 500 nucleotides or longer than 5000 nucleotides were reserved exclusively for testing, while the remaining sequences (with lengths between 500 and 5000 nucleotides) were used

for training. This design simulates real-world scenarios where input sequences frequently deviate from the distributions seen during model development. In total, 510 sequences fell into the atypical category and were held out as the test set, while the rest were used to train the models. Thanks to the careful dataset construction strategy adopted in the earlier stages of our study, both the training and test subsets in this experiment ensured full label coverage.

The goal of this evaluation was to determine whether each method could maintain predictive reliability when applied to samples that significantly deviate from typical input characteristics in terms of length. As shown in Table S4 and Table S5, LncTracker consistently outperformed all baseline methods under this stringent setting. Notably, it achieved the highest scores on four out of six example-based metrics and four out of five label-based metrics. Among these, Average AUC and Average MCC demonstrated particularly strong gains, highlighting the model's effectiveness even in highly imbalanced

multi-label scenarios. This robustness can be attributed to LncTracker’s ability to jointly leverage both sequence-level and structure-level representations, underscoring LncTracker’s strong generalization ability, making it well-suited for real-world applications involving RNA inputs with highly variable lengths.

REFERENCES

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